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Research

Biosynthesis of kratom opioids

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Summary

- Mitragynine, an analgesic alkaloid from the plant *Mitragyna speciosa* (kratom), offers a safer alternative to clinical opioids such as morphine, owing to its more favorable side effect profile. Although kratom has been traditionally used for stimulation and pain management in Southeast Asia, the mitragynine biosynthesis pathway has remained elusive.
- We embarked on a search for mitragynine biosynthetic genes from the transcriptomes of kratom and other members of the Rubiaceae family. We studied their functions *in vitro* and *in vivo*.
- Our investigations led to the identification of several reductases and an enol methyltransferase that forms a new clade within the SABATH methyltransferase family. Furthermore, we discovered a methyltransferase from *Hamelia patens* (firebush), which catalyzes the final step. With the tryptamine 4-hydroxylase from the psychedelic mushroom *Psilocybe cubensis*, we accomplished the four-step biosynthesis for mitragynine and its stereoisomer, speciogynine in both yeast and *Escherichia coli* when supplied with tryptamine and secologanin.
- Although we have yet to pinpoint the authentic hydroxylase and methyltransferase in kratom, our discovery completes the mitragynine biosynthesis. Through these breakthroughs, we achieved the microbial biosynthesis of kratom opioids for the first time. The remarkable enzyme promiscuity suggests the possibility of generating derivatives and analogs of kratom opioids in heterologous systems.

Introduction

Mitragynine from the plant *Mitragyna speciosa* (kratom) and its derivatives and semisynthetic analogs including 7-hydroxymitragynine and mitragynine pseudoindoxyl are new opioids with superior side effect profiles (Kruegel *et al.*, 2016, 2019). Comparing to the widely used opioids such as morphine, these monoterpenoid indole alkaloid (MIA)-based opioids have distinct opioid receptor agonism and signal transduction pathways, which do not recruit β-arrestin-2 that is linked with respiration depression (Takayama, 2004; Kruegel *et al.*, 2016, 2019; Váradi *et al.*, 2016; Bhowmik *et al.*, 2021). In animal models, kratom opioids did not promote self-administration (Yue *et al.*, 2018), suggesting their use to be less addictive compared with opioids responsible for the devastating overdose epidemic around the world. Recent structure activity relationship (SAR) studies have shown that the scaffold of kratom opioids is amenable to modifications and substitutions, which could further improve the safety of these drugs (Kruegel *et al.*, 2019; Bhowmik *et al.*, 2021). While kratom has been historically used as an ethnobotanical remedy for hundreds of years in Southeast Asia and the total synthesis has been reported (Kruegel *et al.*, 2016), the biosynthesis of kratom opioids from the central precursor strictosidine, from which over 3000 MIAs derive, has not been studied.

It is well known that strictosidine deglycosylation leads to numerous unstable strictosidine aglycone structures that exist in spontaneous equilibrium (Fig. 1). The aglycones are further reduced by many iminium reductases to more stable forms responsible for the early diversification of MIA skeletons (Stavrinides *et al.*, 2016; Qu *et al.*, 2017, 2018a). The mitragynine biosynthetic pathway (Fig. 1) from strictosidine likely involves four steps (red arrows in Fig. 1): (a) reduction in strictosidine aglycones to demethylcorynantheidine (**2**); (b) 17-enol methylation of **2** to form corynantheidine (**9**); (c) indole 9-hydroxylation; and (d) 9-*O*-methylation to form mitragynine (**18**).

We reasoned that the strictosidine reduction step is likely catalyzed by an iminium reductase homologous to other characterized reductases such as geissoschizine synthase (GS) and heteroyohimbine synthases (HYS) that also reduce the same substrates (Figs 1, 2a; Stavrinides *et al.*, 2016; Qu *et al.*, 2017, 2018a). The following 17-enol methylation is unique since, to our knowledge, none of the characterized *O*-methyltransferases (OMTs) catalyze nonaromatic enol methylation, suggesting the involvement of a different OMT. While it is common for MIAs to be hydroxylated at C10 or C11 on the indole moiety, the indole C9 hydroxylation is rare and has only been reported in few MIA producing species such as kratom and *Uncaria gambier* (Merlini *et al.*, 1967; Martins &

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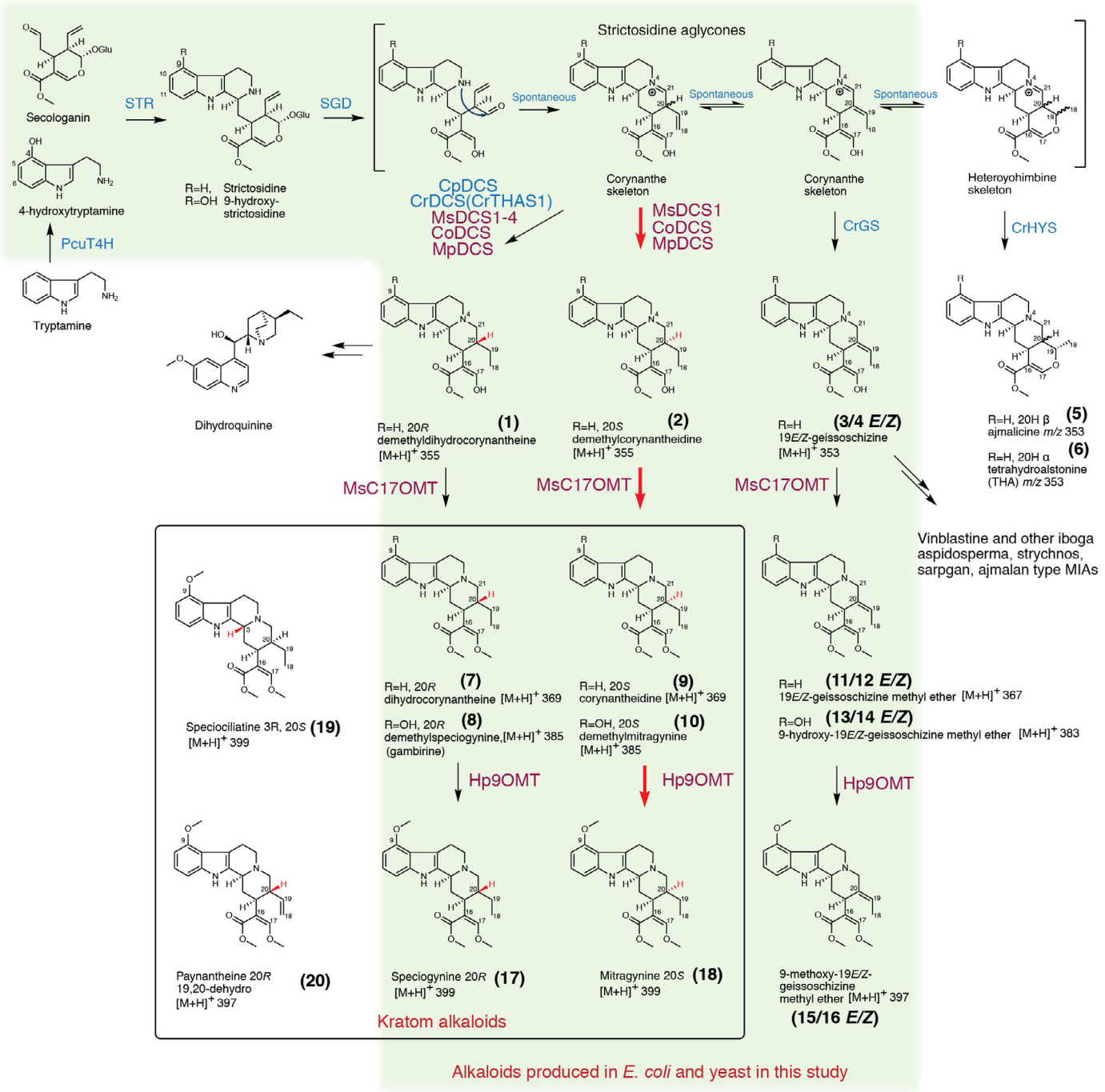


Fig. 1 Biosynthetic pathway for kratom alkaloids and other related monoterpenoid indole alkaloids. The enzymes in dark red are new enzymes identified and characterized in this study. The enzymes in blue are previously characterized enzymes, while new demethyldihydrocorynantheine synthase activity was reported for the *Catharanthus roseus* tetrahydroalstonine synthase 1 (CrTHAS1). The red arrows indicate the biosynthesis of mitragynine (20*S*). The black box shows the naturally accumulating alkaloids in kratom leaves. The alkaloids in green shade are biosynthesized in *Escherichia coli*, yeast, and *in vitro* in this study.

Nunez, 2015). Like MIA C10/C11 hydroxylations, these hydroxyls are often methylated by OMTs, which are commonly found in the cation-independent OMT group that includes many aromatic hydroxyl OMTs such as the MIA 11-hydroxytabersonine OMT (Cr11OMT or Cr16OMT, Fig. 2b; Lam *et al.*, 2007; Murata *et al.*, 2008; Lashley *et al.*, 2022).

In this study, we report the discovery and characterization of nine reductases, a C17-enol OMT, and a 9-OMT that are responsible for the biosynthesis of mitragynine and related corynanthe alkaloids in kratom and other MIA-producing plant species. Among these new enzymes, two of them were also reported in a recent study (Schotte *et al.*, 2023). While completion of the

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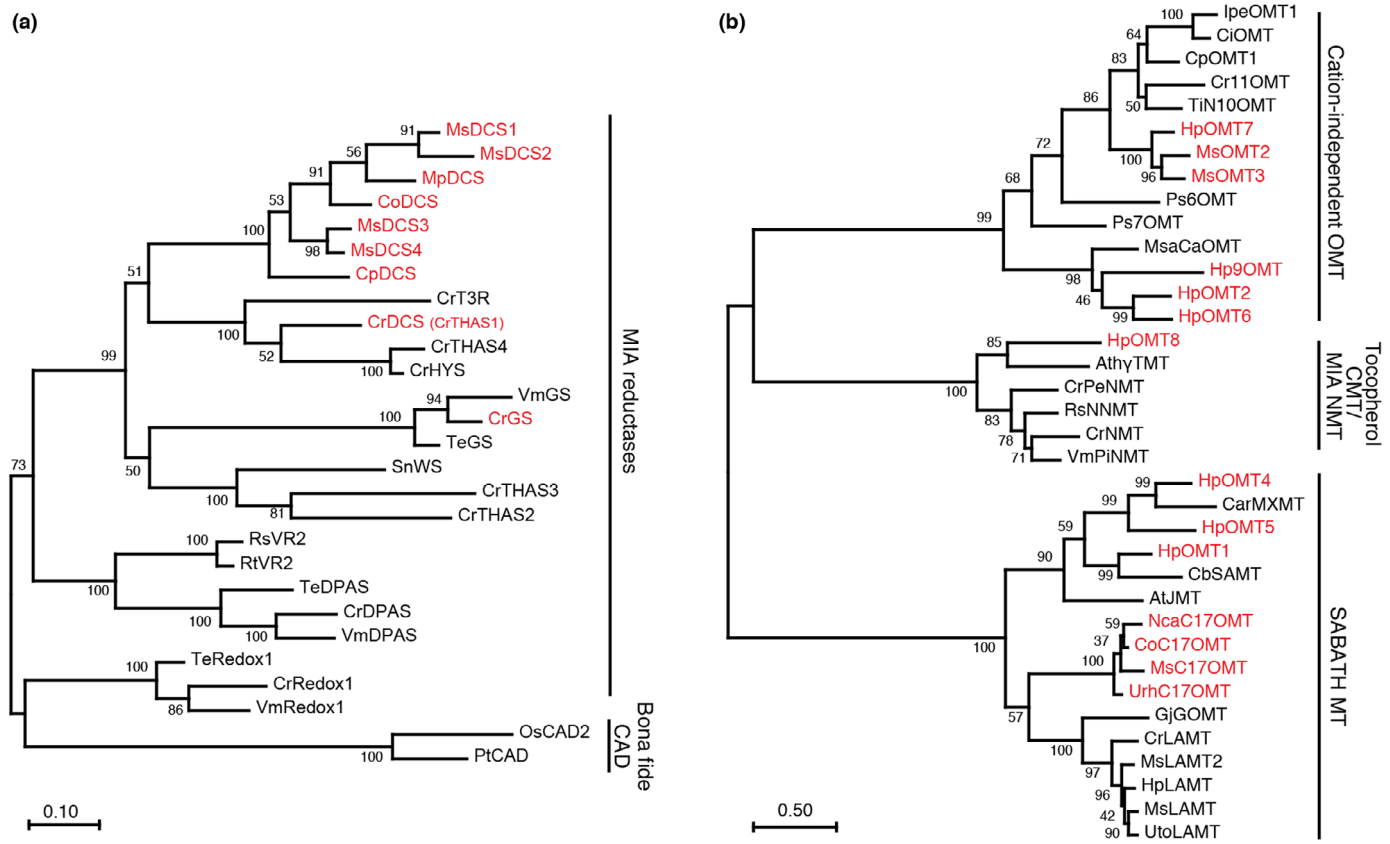


Fig. 2 Phylogenetic analysis of monoterpenoid indole alkaloid (MIA)-producing reductases (a) and methyltransferases (b) in this study. The plant species, amino acid sequences, and Genbank accession numbers are included in Supporting Information Dataset S1. The enzymes labeled in red were investigated in this study. Evolutionary analyses were conducted in MEGA11. The evolutionary history was inferred by using the Maximum Likelihood method and JTT matrix-based model. The percentage of trees in which the associated taxa clustered together is shown next to the branches (100 bootstrap replicates). Initial tree(s) for the heuristic search were obtained automatically by applying Neighbor-Join and BioNJ algorithms to a matrix of pairwise distances estimated using the JTT model, and then selecting the topology with superior log likelihood value. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site (scale bar). CAD, cinnamyl alcohol dehydrogenase; DCS, demethyldihydrocorynantheine/demethylcorynantheidine synthase; DPAS, dihydroprecondylocarpine acetate synthase; GS, geissoschizine synthase; HYS, heteroyohimbine synthase; MT, methyltransferase; THAS, tetrahydroalstonine synthase; VR2, vomilenine reductase 2; WS, Wieland-Gumlich aldehyde synthase.

mitragynine pathway still requires the discovery of the corynantheidine 9-hydroxylase, this study documents the microbial biosynthesis of mitragynine and its epimer speciogynine in *Escherichia coli* using kratom enzymes when supplying the unnatural substrate 4-hydroxytryptamine, which bypasses the need of a hydroxylase. We also show the biosynthesis of mitragynine and speciogynine in *Saccharomyces cerevisiae* (baker's yeast) from tryptamine substrate by including a tryptamine 4-hydroxylase (PcuT4H) from the psychedelic mushroom *Psilocybe cubensis* (Fricke *et al.*, 2017). These results document the first microbial biosynthesis of mitragynine pharmacophore-based analgesics.

Materials and Methods

Chemicals standards

The standards for mitragynine, speciogynine, speciociliatine, paynantheine, corynantheidine, 4-hydroxytryptamine, and 5-hydroxytryptamine were purchased from Cayman Chemicals (Ann Arbor, MI, USA). The standard for 6-hydroxytryptamine was purchased from Toronto Research Chemicals (Toronto, ON, Canada), and the standard for 19*E*-geissoschizine methyl ether was purchased from AvaChem Scientific (San Antonio, TX, USA). The substrate 19*E*-geissoschizine and 19*Z*-geissoschizine was prepared and purified as described previously (Qu *et al.*, 2017). The substrate demethyldihydrocorynantheine was prepared from *in vitro* reaction (10 ml) containing 20 mM Tris–HCl pH 7.5, 1 mM NADPH, 100 μg trptamine, 100 μg secologanin, and purified recombinant proteins: CrSTR (200 μg), CrSGD (20 μg), and MsDCS4 (200 μg). The reaction took place at 30°C for 1 h, and then it was extracted with 10 ml of ethyl acetate. The evaporated extract was reconstituted in methanol and separated by thin-layer chromatography using TLC-silica gel 60 f254 (Sigma-Aldrich) with solvent ethyl acetate: methanol (9 : 1, v/v), which afforded 20 μg demethyldihydrocorynantheine.

Plant materials, crude protein isolation, and RNA/cDNA synthesis

The plants *Mitragyna speciosa* Korth., *Mitragyna parvifolia* (Roxb.) Korth, *Cephalanthus occidentalis* L., *Cinchona pubescens* Vahl, *Catharanthus roseus* (L.) G. Don, and *Hamelia patens* Jacq.

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were grown in a glasshouse at 28°C with 16 h : 8 h, light : dark photoperiod. Leaf tissues (3 g) and 0.2 g polyvinylpolypyrrolidone were ground in liquid nitrogen with mortar and pestle, which were extracted with ice-cold sample buffer (20 mM Tris–HCl pH 7.5, 100 mM NaCl, 10% (v/v) glycerol). The extracts were centrifuged at 15 000 ***g*** for 30 min, and desalted into the same sample buffer with a PD10 desalting column (Cytiva, Marlborough, MA, USA) according to the manufacture's protocol. The total proteins were desalted one more time, and the final samples were stored at −80°C. Leaf tissues (100 mg) were collected for RNA extraction using standard TRIzol RNA isolation reagent according to the manufacture's protocol (ThermoFisher Scientific, Waltham, MA, USA). The resulting RNA was used to generate cDNA using the LunaScript®RT SuperMix Kit according to the manufacture's protocol (New England Biolabs, Ipswich, MA, USA).

Cloning

For His-tagged protein purifications, MsDCS1, MsDCS2, and MpDCS were amplified using the same primer set (1/2) from respective leaf cDNA. MsDCS3 and MsDCS4 were amplified using the same primer set (3/4). CpDCS was amplified using the primer set (5/6). CoDCS was amplified using primer set (7/8). MsC17OMT (MsOMT1), MsOMT2, and MsOMT3 were amplified using primer set (9–14), respectively. HpOMT1-8 were amplified using the primer sets (15–30), respectively. The primers are listed in Supporting Information Table S1. MsOMsDCS1-4, MpDCS, CoDCS, MsC17OMT, and MsOMT2 were cloned in pET30b+ vector within BamHI/SalI sites. MsOMT3 was cloned in pET30b+ vector within SalI/NotI sites. HpOMT4 was cloned in pRSF-duet vector within SacI/NotI site. All remaining HpOMTs were cloned in pRSF-duet vector within SalI/NotI site. All these vectors were mobilized to *E. coli* BL21DE3 for expression. CpDCS were gateway-cloned into pDEST17 vector by Gateway™ BP and LR clonase™ II Enzyme mix according to the manufacturer's protocol (ThermoFisher Scientific), and mobilized to *E. coli* BL21A1 for expression. The Genbank nos. for MsDCS1-4, Ms/Co/Nca/UrhC17OMT, and Hp9OMT are OQ129427–129436, and OQ515471–515478.

For pathway assembly in *E. coli*, N-terminal truncated CrSTR was subcloned into both pACE and pDC vectors (MultiColi™ system; Geneva Biotech, Pregny-Chambésy, Switzerland) by releasing the gene with XbaI/XhoI from pET30b + dCrSTR previously described (Qu *et al.*, 2017) and ligating it within the same sites. CrSGD was subcloned into pDS vector by releasing the gene with XbaI/XhoI from pET30b + CrSGD (Qu *et al.*, 2017) and ligating it within the same sites. CrGS was subcloned into pACE vector by releasing the gene with BamHI/XhoI from pET30b + CrGS vector (Qu *et al.*, 2017) and ligating it within the same sites. The triple expression vector containing CrdSTR, CrSGD, and one of the CrGS, MsDCS1, MsDCS4 was created by Cre-recombinase (New England Biolabs) according to the manufacturer's protocols (Geneva Biotech). MsDCS1 and MsDCS4 were subcloned from respective pET30b+ vectors using BamHI/XhoI into pDC vectors. MsC17OMT was subcloned to pRSF-duet vector by releasing it with NdeI/SalI from the respective pET30b+ vector and ligating it within NdeI/XhoI sites. Into the same pRSF-duet vector, Hp9OMT was subcloned by SalI/NotI. The biosynthetic pathway was constructed by cotransforming the above vectors into *E. coli* BL21A1 cells.

For pathway assembly in yeast, codon-optimized dCrSTR (Shahsavarani *et al.*, 2022) was subcloned from pUG19 vector into pESC-His vector within NotI/ClaI site. MsDCS1 was subcloned from the respective pET30b+ vector into pESC-His-dCrSTR within the BamHI/SalI sites. The expression cassette of codon-optimized CrSGD was subcloned from pESC-His-CrSGD vector (Gao *et al.*, 2022) by the primer set (31/32), and ligated into pESC-His-dCrSTR-MsDCS1 vector within the DraIII site. The PcuT4H open reading frame was synthesized (Twist Biosciences, South San Francisco, CA, USA) and subcloned into pESC-Leu-CrCPR vector (Shahsavarani *et al.*, 2022) within the BamHI/SalI sites. MsOMT1 was subcloned from the respective pET30b+ vector into pESC-Ura vector within BamHI/SalI sites. Hp9OMT was amplified by the primer set (33/34) and cloned into pESC-Ura-MsOMT1 vector within NotI/SpeI sites. The vectors were mobilized into yeast strain BY4747 (MATα his3Δ1 leu2Δ0 met15Δ0 ura3Δ0 YPL154c::kanMX4; Qu *et al.*, 2018a).

Recombinant protein expression and purifications

An overnight culture (2 ml) of *E. coli* BL21DE3 strains containing MsDCS1-4, MpDCS, MsC17OMT, MsOMT2, and MsOMT3 in pET30b+ vectors were used to inoculate 200 ml of LB media, which were cultured at 200 rpm and 37°C until OD600 reached 0.6–0.7. The cultures were induced with 0.1 mM IPTG at 15°C, 200 rpm overnight. For CpDCS and CoDCS, an overnight culture (2 ml) of *E. coli* BL21A1 was used to inoculate 200 ml of LB media, which were cultured at 200 rpm and 37°C until OD600 reached 0.3. The cultures were induced with 0.1% (w/v) galactose at 15°C, 200 rpm overnight. The induced cultures were sonicated in ice-cold sample buffer (20 mM Tris–HCl pH 7.5, 100 mM NaCl, 10% (v/v) glycerol) and purified using standard Ni-NTA affinity chromatography. After eluting with 250 mM imidazole in sample buffer, the purified recombinant proteins were desalted using a PD-10 desalting column (Cytiva) according to the manufacturer's protocol into the same sample buffer and stored at −80°C.

In vitro assays and kinetics

A standard *in vitro* reaction (50 µl) included 20 mM Tris–HCl pH 7.5, 1 mM NADPH, and some or all these components: 60 µM SAM, 2 µg CrSTR, 0.2 µg CrSGD, 2 µg MsDCS1-4, CoDCS, MpDCS, CrGS, or CrDCS(CrTHAS1), 2 µg MsC17OMT, and 2 µg Hp9OMT. The substrates included 1 µg secologanine and 1 µg tryptamine or its hydroxylated derivatives. The reaction was incubated at 30°C for 1 h and terminated by adding 150 µl methanol. The kinetics triplicated assays (50 µl) included 20 mM Tris–HCl pH 7.5, 100 µM SAM, 1 µg MsC17OMT, and substrate 19*E*-, or 19*Z*-geissoschizine, or demethyldihydrocorynantheine at 1.5, 2.5, 4, 12, 24, 36, 60, and

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100 μM concentrations. The kinetics assays were performed at 30°C for 2 min before they were terminated by adding 150 μl methanol to the reactions. The products were quantified using a standard curve, to generate the enzyme velocity. The kinetics parameters and saturation curves were approximated using the software PRISM 9.5.0 (GraphPad Software, LLC., Boston, MA, USA).

In vivo biotransformation

Escherichia coli strain BL21A1 containing various biosynthetic pathway genes were inoculated in 1 ml of LB media with appropriate antibiotics and 2% glucose (w/v) overnight at 37°C in a shaking incubator. The co-expression of CrSTR and CrSGD significantly impacted the *E. coli* growth and 2% glucose (w/v) sufficiently inhibited leaky protein expression in uninduced BL21A1 cells. The overnight cultures were used to inoculate 10 ml of fresh LB media (1 in 100 dilution) with appropriate antibiotics and 2% glucose (w/v), which were further grown at 37°C in a shaking incubator until OD_{600} reached 1.0. The cells were then collected by centrifugation, washed once with water, and resuspended in 10 ml of fresh LB media with appropriate antibiotics and 0.1% arabinose (w/v) to induce protein expression. The cultures were incubated in a shaking incubator at 15°C overnight. The induced cells were collected and resuspended in 2 ml Tris HCl pH 7.5 supplemented with 10% (v/v) LB broth. The substrates secologanin (10 μg), tryptamine (10 μg), and/or 4-hydroxytryptamine (10 μg) were added to the biotranformation mixture, which was incubated in a shaking incubator at 30°C for MsDCS1/4 and 15°C for CrGS for 24 h. The cultures were mixed with equal volume of methanol and used for liquid chromatography tandem mass spectrometry (LC–MS/MS) analyses.

Yeast strain BY4747 (MATα his3Δ1 leu2Δ0 met15Δ0 ura3Δ0 YPL154c::kanMX4; Qu *et al.*, 2018a) containing various biosynthetic genes were inoculated in 1 ml of standard synthetic complete (SC) media with 2% glucose (w/v) overnight at 30°C in a shaking incubator. The cells were collected by centrifugation, washed once with water, and resuspended in 1 ml of SC media with 2% galactose (w/v) for 24 h at 30°C in a shaking incubator. The induced cells were collected and resuspended in 1 ml Tris HCl pH 7.5 with the substrates secologanin (10 μg), tryptamine (10 μg), and/or 4-hydroxytryptamine (10 μg). The biotransformation mixtures were incubated in a shaking incubator at 30°C overnight, and then equal volume of methanol was added for LC–MS/MS analyses.

LC–MS/MS

Liquid chromatography tandem mass spectrometry was performed on an Agilent Ultivo Triple Quadrupole LC–MS equipped with an Avantor® ACE® UltraCore™ SuperC18™ column (2.5 μm, 50 × 3 mm), which included the solvent systems: solvent A, methanol : acetonitrile : ammonium acetate 1 M : water at 29 : 71 : 2 : 398; solvent B, methanol : acetonitrile : ammonium acetate 1 M : water at 130 : 320 : 0.25 : 49.7. The following linear gradient (8 min, 0.6 ml min^{-1}) were used: 0 min 80% A, 20% B; 0.5 min, 80% A, 20%B; 5.5 min 1% A, 99% B; 5.8 min 1% A, 99% B; 6.5 min 80% A, 20% B; 8 min 80% A, 20% B. The photodiode array detector records from 200 to 500 nm. The MS/MS was operated with gas temperature at 300°C, gas flow of 10 l min^{-1}, capillary voltage 4 kV, fragmentor 135 V, collision energy 30 V with positive polarity. The Qualitative Analysis 10.0 software by Agilent was used for all LC analyses.

Results

Discoveries of four demethyldihydrocoryanthine/demethylcorynantheidine synthases in kratom

Previously, our group and others characterized the key reductase geissoschizine synthase (CrGS) from the plant *Catharanthus roseus* (Madagascar's periwinkle; Qu *et al.*, 2017, 2018a; Tatsis *et al.*, 2017), which reduces strictosidine aglycones to corynanthe type MIAs 19*E*- and 19*Z*-geissoschizine ($[M+H]^+$ *m/z* 353, 19,20-dehydro, **3**, **4**). The 19*E* intermediate is the precursor to major strychnos, sarpagan, iboga, and aspidosperma MIA skeletons (Fig. 1). In addition to geissoschizine, the heteroyohimbine MIAs such as ajmalicine (**5**), tetrahydroalstonine (THA, **6**), and mayumbine are also derived from reducing strictosidine aglycones by homologous reductases such as heteroyohimbine synthase (CrHYS) and tetrahydroalstonine synthase (CrTHAS1-4; Stavrinides *et al.*, 2016). Studies with *Cinchona pubescens* showed that another reductase, demethyldihydrocorynantheine synthase (CpDCS; Trenti *et al.*, 2021), reduced strictosidine aglycones to demethyldihydrocorynantheine (*m/z* 355, corynanthe type, 20*R*, **1**), the precursor to the antimalarial drug quinine (Fig. 1). By contrast, the biosynthesis of mitragynine (**18**) involves the formation of demethylcorynantheidine (**2**), the 20*S*-epimer of demethyldihydrocorynantheine (**1**). In addition to mitragynine and its intermediates, kratom and other related Rubiaceae species also make several stereo- and geometric isomers differing at C3 and C18-19-20 (Fig. 1; Takayama, 2004; Martins & Nunez, 2015). These diverse structures, such as specioginine (20*R*, **17**), speciociliatine (3*R*, **19**), and paynantheine (20*R*, 18,19-dehydro, **20**), suggest that different reductases are likely involved in their biosynthesis.

Using the sequences of CrGS, CrTHAS1, and CpDCS for comparison led to the identification and cloning of four homologous reductases from the *M. speciosa* leaf transcriptomes (http://mpgr.uga.edu) named demethyldihydrocorynantheine/demethylcorynantheidine synthase 1–4 (MsDCS1-4). While MsDCS1 was cloned by Trenti *et al.* (2021; Genbank MW456555), the gene product and its biochemical function were not characterized. The four MsDCS shared 80–95% amino acid sequence identity and were clustered together with other characterized MIA iminium reductases and short/medium chain alcohol dehydrogenases in a phylogenetic analysis (Fig. 2a).

We expressed and purified *N*-terminal His-tagged MsDCS1-4 in *E. coli* (Fig. S1) and tested their biochemical activities *in vitro*. The strictosidine aglycones (*m/z* 351) were produced by the *C. roseus* strictosidine synthase (CrSTR) and strictosidine β-glucosidase (CrSGD) using the substrate tryptamine and

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secologanin (Fig. 1), which were coupled to MsDCS *in vitro* reactions. We also compared these reactions to those with the characterized enzymes CrGS, CrTHAS1, and CpDCS. As expected from previous studies, CrGS reduced strictosidine aglycones (*m/z* 351) to produce both 19*E*- and 19*Z*-geissoschizine (*m/z* 353, **3, 4**) with THA (*m/z* 353, **6**) by-product (Fig. 3a). As expected from previous studies, CpDCS produced a major product of demethyldihydrocorynantheine (*m/z* 355, 20*R*, **1**), which is intermediate to mitragynine 20*R*-epimer speciogynine (**17**) (Fig. 3a). Several isomeric MIAs (*m/z* 353 and *m/z* 355) were also produced as minor products in these reactions. We also noticed that CrTHAS1 produced not only THA but also significant amounts of **1**, which has not been reported (Fig. 3a). Like CpDCS, MsDCS2-4 produced **1** as a major product together with THA (**6**) (*m/z* 353) and other minor (*m/z* 355) products (Fig. 3). By contrast, a major MsDCS1 product was a new *m/z* 355 MIA (Fig. 3a,b) with an elution time different from that of **1**. In the same reaction, MsDCS1 also produced 19*Z*-geissoschizine (*m/z* 353, **4**), THA (**6**), yohimbine (*m/z* 355), and several other minor *m/z* 353 MIA products when compared to authentic standards (Fig. 3b). The Electrospray Ionization tandem Mass Spectrometry (ESI-MS/MS) fragment patterns of this new *m/z* 355 MIA was nearly identical to that of **1** (Fig. 3c). We later identified this MIA as demethylcorynantheidine (20*S*, **2**), because it was enzymatically converted to corynantheidine (**9**) and mitragynine (**18**) by the two methyltransferases discovered in this study. The C17-enol methylation on DCS and GS products dramatically improved their poor chromatograms (wide peaks), which allowed us to properly identify the methylated derivatives with authentic standards (Fig. 3a, panel 2–4). The ESI-MS/MS fragment patterns of standards, known MIA products from enzymatic reactions, and MIAs produced in this study are also included in Fig. S2.

Discovery of the corynanthe C17-enol OMT and two additional demethyldihydrocoryanthine/demethylcorynantheidine synthases

With these findings, we continued to discover the methyltransferase (MT) responsible for 17-enol methylation. To our knowledge, no MTs discovered to date methylate a nonaromatic enol oxygen. The well-studied plant MTs include the SABATH (Salicylic acid carboxyl methyltransferase, Benzoic acid carboxyl MT, and Theobromine synthase) MTs that methylate carboxylic acids such as salicylic acid and jasmonic acid (Zubieta *et al.*, 2003). The loganic acid MT (LAMT) in *C. roseus* involved in MIA biosynthesis is also a member of this family (Murata *et al.*, 2008). Other characterized MTs include the aromatic *O*-methyltransferases (OMT) that methylate the aromatic hydroxyls such as flavonoids and benzylisoquinoline alkaloids, steroid *C*-methyltransferases (CMT) that methylate many triterpenoids, and the tocopherol CMTs, some of which have evolved to perform MIA *N*-methylation (Lam *et al.*, 2007; Levac *et al.*, 2008; Liscombe *et al.*, 2010; Cázares-Flores *et al.*, 2016; Fig. 2b).

Using the sequences of *C. roseus* CrLAMT (SABATH type) and 11-hydroxytabersonine OMT (Cr11OMT, also known as Cr16OMT, aromatic OMT) in MIA biosynthesis, we identified three candidate MTs from the kratom transcriptome, which included a SABATH-type MsOMT1 and two homologous (70% amino acid sequence identity) aromatic OMT type: MsOMT2 and 3. We further identified other MsOMT1 homologs from the transcriptomes of *Uncaria rhynchophylla* (cat's claw), *Nauclea cadamba* (burflower tree), and *Cephalanthus occidentalis* (buttonbush), which all produce MIAs and reside in the Naucleeae tribe of the Rubiaceae family. No MsOMT1 homologs of > 55% identity could be identified outside this tribe when searching in National Center for Biotechnology Information (NCBI) total nonredundant protein database, nor could such homologs be identified in another Rubiaceae species *Cinchona pubescens* in the tribe Cinchoneae or in MIA producing species *C. roseus*, *Vinca minor*, and *Tabernaemontana elegans* in the Apocynaceae family. The phylogenetic analysis (Fig. 2b) showed that MsOMT1 and its Naucleeae homologs form a distinct clade within the SABATH family, and they are clearly distinguished from other OMTs where MsOMT2 and 3 locate.

We expressed and purified *N*-terminal His-tagged MsOMT1-3 in *E. coli*, and tested their activities using purified 19*E*-, 19*Z*-geissoschizine (**3**, **4**), and demethyldihydrocorynantheine (20*R*, **1**). Clearly MsOMT1 was the expected C17-enol OMT, as it methylated (gain of 14 amu) all three substrates to respective geissoschizine methyl ethers (**11**, **12**) and dihydrocorynantheine (**7**). The identity of 19*E*-geissoschizine methyl ether was also supported by comparing it to a commercial standard. MsOMT1 showed typical Michaelis–Menten enzyme kinetics for all three substrates (K_M 5.3–28.7 μM, Fig. S3). MsOMT1 did not show activity with other tested MIAs including reserpic acid, yohimbic acid, yohimbine, corynanthine, 3-hydroxy-2,3-dihydrotabersonine, 11-hydroxytabersonine, vincamine and ajmaline, or simple phenolic acids including gallic acid, syringic acid, salicylic acid, benzoic acid, caffeic acid, *p*-coumaric acid, ferulic acid and trans-cinnamic acid. The results suggested that MsOMT1 likely only accepts corynanthe MIA with a free 17-enol. We therefore renamed MsOMT1 as corynanthe 17-OMT (MsC17OMT). In comparison, MsOMT2 and 3 did not accept any tested MIAs as substrates.

With MsC17OMT, we revisited the activities of MsDCS1-4 in coupled reactions involving these enzymes. With MsC17OMT, the methylated dihydrocorynantheine (**7**) formed at the expense of demethyldihydrocorynantheine (20*R*, **1**) in MsDCS2-4 reactions (Figs 3a, S4). With MsC17OMT, the reaction involving MsDCS1 clearly showed three major products: corynantheidine (20*S*, **9**) that is intermediate to mitragynine (**18**), dihydrocorynantheine (**7**), and 19*Z*-geissoschizine methyl ether (**12**) (Fig. 3a). Owing to the 17-*O*-methylation, the methylated products could no longer exist in enol-keto resonance, and they elute as sharp peaks during LC separation. The improved liquid chromatograms and availability of authentic corynantheidine, 19*E*-geissoschizine methyl ether standards allowed confident compound identification (Fig. 3a,c). The identification of corynantheidine (**9**) by MsDCS1 and MsC17OMT confirmed our previous identification of demethylcorynantheidine (**2**) when MsC17OMT was not involved. With this finding, we cloned two more homologous

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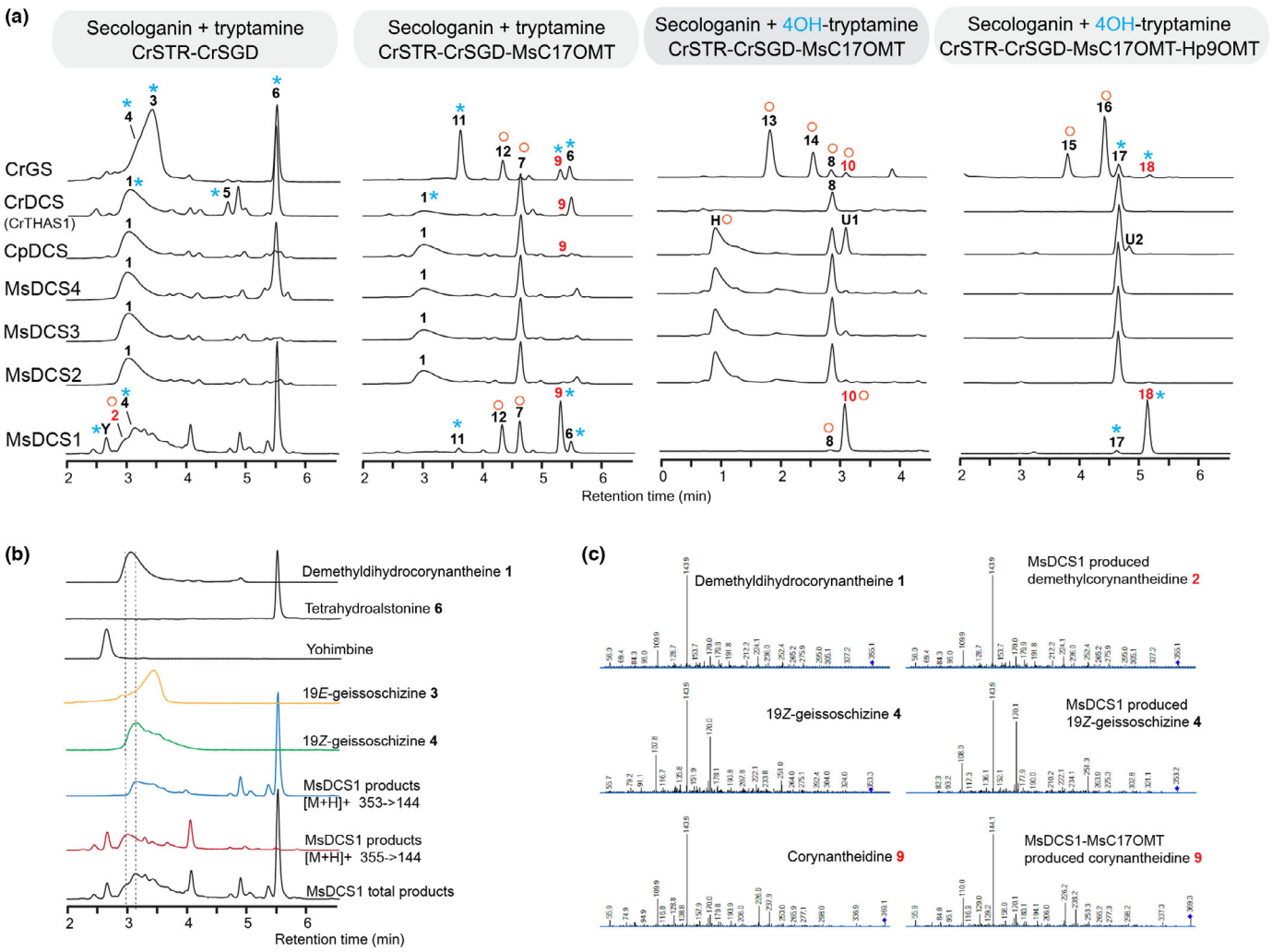


Fig. 3 Biosynthesis of mitragnine, speciogynine, 9-methoxygeissoschizine methyl ether epimers by recombinant enzymes. (a) Liquid chromatography tandem mass spectrometry (LC–MS/MS) chromatograms of coupled enzyme activities from CrSTR, CrSGD, CrGS, CrDCS (CrTHAS1), CpDCS, MsDCS1-4, MsC17OMT (MsOMT1), and Hp9OMT (HpOMT3) with secologanin and tryptamine/4-hydroxytryptamine substrates. The structures of peaks with numeric numbers can be found in Fig. 1. The peaks labeled with red numbers indicate that they are intermediate to mitragynine. The blue stars indicate that the alkaloids were confirmed with authentic standards, while the orange circles indicate that the alkaloids were either enzymatically converted to authentic standards or enzymatically converted from authentic standards. Unless noted, all LC–MS/MS chromatograms are shown by combining these ion transitions together: electrospray ionization tandem mass spectrometry $[M+H]^+$ *m/z* 353➔144, 355➔144, 367➔144, 369➔144, 383➔160, 385➔160, 397➔174, and 399➔174. Peak Y, yohimbine; Peak H, 9-hydroxy-demethyldihydrocorynantheine; Peak U1, unknown monoterpenoid indole alkaloids (MIAs) with $[M+H]^+$ 383; Peak U2, $[M+H]^+$ 397 (383 + 14, methylation). (b) Comparing the LC–MS/MS chromatograms of MsDCS1 products to authentic standards. (c) Comparing the MS/MS product ion fingerprints of MsDCS1 and MsC17OMT products to authentic standards.

reductases CoDCS and MpDCS from the plants *Cephalanthus occidentalis* and *Mitragyna parvifolia*. When assayed in the same coupled reactions with MsC17OMT, both CoDCS and MpDCS generated two products, dihydrocorynantheine (**7**) and corynantheidine (**9**) (Fig. S4). It is also worth noting that the previously characterized reductases CrGS, CrTHAS1, and CpDCS also produced various amounts of corynantheidine (20*S*, **9**) in these MsC17OMT-coupled assays (Fig. 3a), which could not be easily identified without 17-*O*-methylation due to the overlapping LC chromatograms of the nonmethylated MIAs. These results indicated the plasticity of strictosidine aglycone reduction by these homologous reductases, which is key to the earlier diversification of MIA skeletons.

Using 4-hydroxytrypamine substrate allowed the discovery of the 9-hydroxyl OMT from firebush

The remaining steps of mitragynine biosynthesis include 9-hydroxylation and 9-*O*-methylation. We reasoned that the 9-hydroxyl group could be artificially introduced earlier in the biosynthesis by using 4-hydroxytryptamine as a substrate based on the promiscuity of these biosynthetic enzymes. Despite the reduced activities, purified recombinant CrSTR, CrSGD, MsDCS1, and MsC17OMT converted secologanin and 4-hydroxytryptamine to demethylmitragynine (*m/z* 385, 20*S*, **10**) and demethylspeciogynine (also known as gambirine, *m/z* 385, 20*R*, **8**) epimers (Fig. 3a). We obtained similar results when we

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replaced MsDCS1 with MpDCS or CoDCS, while the assays containing MsDCS2-4, CrTHAS, and CpDCS only made demethylspeciogynine (*m/z* 385, 20*R*, **8**) as a major product (Figs 3a, S4). Interestingly, CpDCS also produced a less reduced MIA (*m/z* 383, peak U1 in Fig. 3a) that we have not identified. When we swapped MsDCS1 with GS, the reaction led to the formations of 9-hydroxygeissoschizine 19-*E*/*Z* epimers (**13**, **14**) that have not been reported in nature, as well as small amounts of both demethylmitragynine and demethylspeciogynine (Fig. 3a). When we swapped the 4-hydroxytryptamine with 5, or 6-hydroxylated tryptamine in the same reactions, we detected the formation of 10, or 11-hydroxylated isomers of demethylmitragynine and demethylspeciogynine (Fig. S5), suggesting that both the reductases and MsC17OMT tolerate indole hydroxylation at these positions.

To confirm the identification of demethylmitragynine (**10**) and demethylspeciogynine (**8**) from our *in vitro* reactions, we subjected these *in vitro* produced MIAs to the total leaf proteins isolated from kratom, buttonbush, *C. roseus*, and *Hamelia patens* (firebush), which were also supplemented with the methyl donor *S*-Adenosyl methionine (SAM). While the leaf proteins of kratom, buttonbush, or *C. roseus* did not catalyze the final 9-*O*-methylation, firebush leaf proteins converted demethylmitragynine and demethylspeciogynine to the final products mitragynine (**18**) and speciogynine (**17**) that accumulate naturally in kratom leaves, when compared to commercial standards (Fig. 4). The results confirmed the identification of other MIA intermediates and indicated that the final methylation is SAM-dependent.

After searching the firebush leaf transcriptome with sequences of CrLAMT and Cr11OMT, we identified and cloned eight OMT candidates that were named as HpOMT1-8 with decreasing expression levels (Table S2). By feeding the demethylmitragynine (**10**) and demethylspeciogynine (**8**) substrate mixtures to *E. coli* expressing each HpOMT, we discovered that only HpOMT3 methylated both substrates (Fig. S6), which was therefore renamed as Hp9OMT. We purified *N*-terminal His-tagged Hp9OMT (Fig. S1), and included it in the *in vitro* reactions containing CrSTR, CrSGD, various DCS/GS, MsC17OMT, and the substrates 4-hydroxytryptamine and secologanin. As hoped the addition of Hp9OMT to the coupled reaction containing MsDCS1 generated mitragynine and speciogynine rather than demethylmitragynine and demethylspeciogynine (Fig. 3a). Similarly, the coupled reactions generated speciogynine as the major products for MsDCS2-4, CpDCS, and CrTHAS with Hp9OMT, while both mitragynine and speciogynine were formed with CoDCS and CpDCS (Figs 3a, S4). The reaction containing CrGS and Hp9OMT also formed small amounts of mitragynine and speciogynine, but the 9-methoxygeissoschizine methyl ether 19-epimers were the dominant products (Fig. 3a).

In addition to its corynanthe MIA 9-hydroxylation activity, Hp9OMT also accepted 10-hydroxylated corynanthe MIAs as substrates. When we replaced 4-hydroxytryptamine with 5-hydroxytryptamine in the same coupled reactions, all corresponding MIAs were methylated by Hp9OMT (Fig. S5). By contrast, Hp9OMT did not methylate the 11-hydroxylated corynanthe

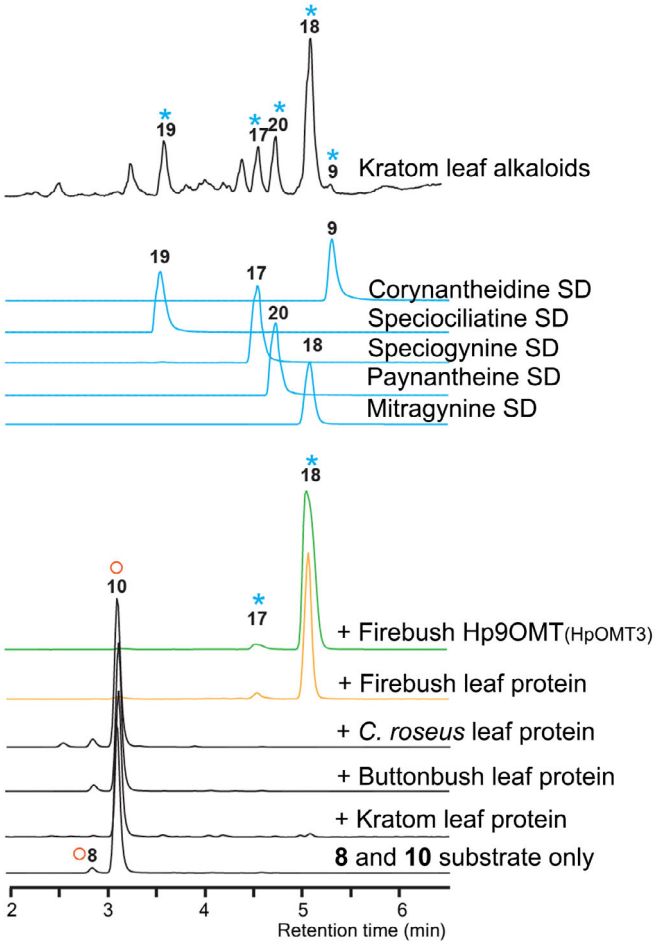


Fig. 4 Demethylmitragynine (**10**) and demethylspeciogynine (**8**) were methylated to form mitragynine (**18**) and speciogynine (**17**) by Hp9OMT (HpOMT3) and the total leaf proteins of *Hamelia patens* (firebush). The blue stars indicate that the alkaloids were confirmed with authentic standards, while the orange circles indicate that the alkaloids were either converted to authentic standards or enzymatically converted from authentic standards. Except for kratom leaf alkaloids that were recorded with a total ion chromatogram, all remaining liquid chromatography tandem mass spectrometry (LC–MS/MS) chromatograms are shown by combining these ion transitions together: electrospray ionization tandem mass spectrometry $[M+H]^+$ *m/z* 353➔144, 355➔144, 367➔144, 369➔144, 383➔160, 385➔160, 397➔174, and 399➔174. Five alkaloid standards were used to identify *in vitro* enzyme products and the total kratom leaf alkaloids in these experiments. The structures can be found in Fig. 1.

MIAs when we used 6-hydroxytryptamine substrate. Therefore, Hp9OMT activity is restricted to C9 and C10 positions.

In vivo biosynthesis of mitragynine in *E. coli* and yeast

The discoveries and characterizations of DCSs, C17OMT, and 9OMT enzymes allowed us to assemble the almost complete mitragynine biosynthetic pathway in both *E. coli* and *Saccharomyces cerevisiae* (baker's yeast). We co-expressed five genes CrSTR, CrSGD, MsDCS1, MsC17OMT, and HpOMT9 on plasmids in *E. coli* and fed the culture with 4-hydroxytryptamine

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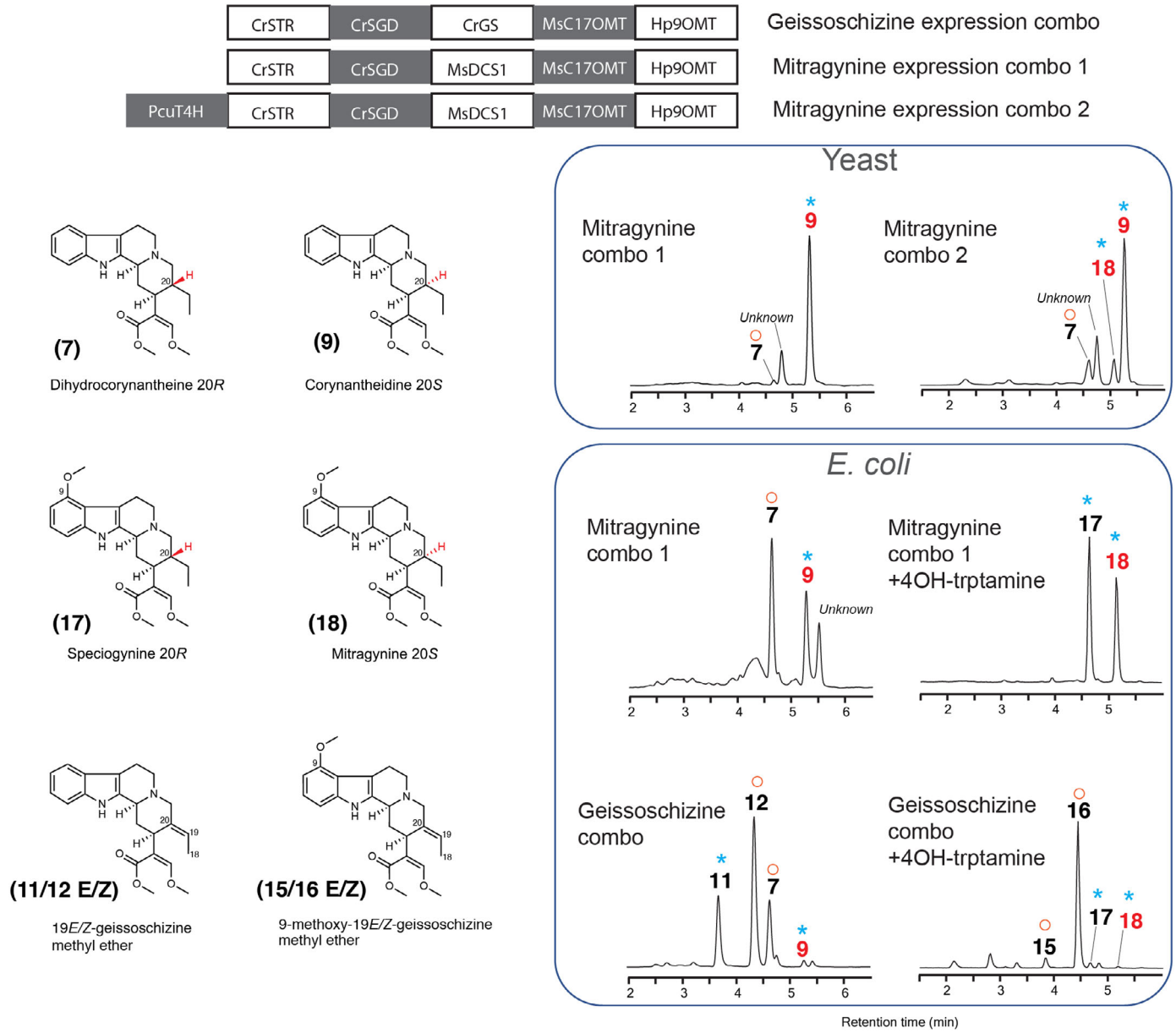


Fig. 5 Biosynthesis of kratom alkaloids mitragynine, speciogynine, and unnatural alkaloid 9-methoxylgeissoschizine methyl ether epimers in *Escherichia coli* and *Saccharomyces cerevisiae* (baker's yeast). Five enzymes CrSTR, CrSGD, MsDCS1 (or CrGS), MsC17OMT, and Hp9OMT were co-expressed in *E. coli*, while the same five enzymes with an additional PcuT4H were co-expressed in yeast. The chromatograms show the alkaloid biosynthesis in culture media after feeding secologanin and either tryptamine or 4-hydroxytryptamine to the cultures. Red numbers indicate that the monoterpenoid indole alkaloids (MIAs) are intermediate to mitragynine. The blue stars indicate that the alkaloids were confirmed with authentic standards, while the orange circles indicate that the alkaloids were either converted to authentic standards or enzymatically converted from authentic standards. All liquid chromatography tandem mass spectrometry (LC-MS/MS) chromatograms are shown by combining these ion transitions together: electrospray ionization tandem mass spectrometry *m/z* $[M+H]^+$ 353➔144, 355➔144, 367➔144, 369➔144, 383➔160, 385➔160, 397➔174, and 399➔174. The unknown peaks all have *m/z* 367.

and secologanin substrates. After overnight incubation, the culture successfully converted the substrates to both mitragynine (18.7 ng ml^{-1}) and speciogynine (24.7 ng ml^{-1}) with minimal by-products (Figs 5, S7). The result again confirmed our identifications of all these enzymes, the intermediates, and their stereochemistry. Replacing MsDCS1 with MsDCS4 led to the sole accumulation of speciogynine (32.9 ng ml^{-1}, Fig. S7), similar to the results from the *in vitro* reactions (Fig. 3a). Replacing MsDCS1 with CrGS mostly resulted in the biosynthesis of 9-methoxy-19*Z*-geissoschizine methyl ether (**16**, 16.7 ng ml^{-1}), and minimal amounts of other MIAs (Figs 5, S7). The opioid activities of 9-methoxygeissoschizine methyl ethers are not known since they are neither natural products nor synthesized for pharmacology studies. Removing Hp9OMT expression in *E. coli* instead led to the biosynthesis of the nonmethylated, 9-hydroxyl versions of respective MIAs for all three reductases (Fig. S7), and feeding tryptamine also led to the formations of the non-hydroxylated MIAs (Figs 5, S7).

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While the corynantheidine 9-hydroxylase remains to be discovered, we attempted the biosynthesis of mitragynine in yeast by using a recently characterized tryptamine 4-hydroxylase (PcuT4H, *pcuH*, genbank MF000993) from the psychedelic mushroom *Psilocybe cubensis* (Fricke *et al.*, 2017), which would supply 4-hydroxytryptamine. Yeast co-expressing the six enzymes CrSTR, CrSGD, MsDCS1, MsC17OMT, HpOMT9, and PcuT4H successfully produced mitragynine (145 ng ml^{-1}) from tryptamine and secologanin substrates, although corynantheidine accumulated as the major product (1190 ng ml^{-1}, 1 : 8 ratio, Fig. 5). Removing PcuT4H from yeast co-expression led to the biosynthesis of corynantheidine as the major product, since no tryptamine hydroxylation occurred (Fig. 5). In both cases, a minor unknown MIA (*m/z* 367) also accumulated, which might be a less reduced version of corynantheidine (*m/z* 369) or its isomer. It is also worth noting that the *in vivo* stereo-specificity of MsDCS1 when expressed in *E. coli* and yeast were different. The MsDCS1-bacteria generated equal levels of 20*S* and 20*R* MIAs, while the 20*S* MIAs (corynantheidine, mitragynine) were dominant in yeast (Fig. 5). In comparison, MsDCS1 produced mostly 20*S* stereochemistry *in vitro* (Fig. 3a,b).

Discussion

Kratom alkaloids such as mitragynine, 7-hydroxymitragynine, and derivatives appear to cause significantly fewer side effects and may have superior safety profiles (Takayama, 2004; Kruegel *et al.*, 2016, 2019; Váradi *et al.*, 2016; Bhowmik *et al.*, 2021). It is interesting that a pharmacophore based on corynanthe MIA skeleton can induce agonism with several opioid receptors, and such a pharmacophore is amenable to further modifications that may lead to even safer opioids (Bhowmik *et al.*, 2021). Structure activity relationship studies showed that the 9-hydroxyl, 17-methoxyl, and the stereochemistry of C15 and C20 are all required for opioid receptor activation. For example, switching the C20 stereochemistry from *S* (mitragynine) to *R* (speciogynine) causes significant loss of opioid activity (Kruegel *et al.*, 2016), indicating the importance of this position with respect to overall receptor interaction. Therefore, it is critical to identify biosynthetic steps responsible for these structural features. The biosynthesis of kratom opioids described here complements the recent invention of their total organic synthesis and reveals an alternative method for acquiring these promising pharmaceuticals that may be developed commercially to mitigate the opioid overdose epidemics around the world.

In this study, we identified six homologous reductases that reduce strictosidine aglycones to the corynantheidine (20*S*) and dihydrocorynantheine (20*R*) skeletons and compared their biochemical properties with three already characterized MIA reductases. The results show their instrumental roles in forming the signature chemical structures required for kratom's opioid activities. The identifications of the demethyldihydrocorynantheine/demethylcorynantheidine synthases (DCSs) were based on previous characterizations of these short/medium chain dehydrogenase type reductases from *C. roseus* and *Cinchona pubescens* (Fig. 2a; Qu *et al.*, 2015, 2017, 2018a,b; Stavrinides *et al.*, 2016; Tatsis *et al.*, 2017; Eng *et al.*, 2022). The recombinant enzymes clearly showed great plasticity in their reduction product spectra since every enzyme produced multiple products, comparable to the well-known structural diversity of the strictosidine aglycones that exist in equilibrium (Fig. 1). Such product diversity was not easily discerned due to the low abundance of some products and their overlapping liquid chromatography behaviors. Remarkably, the C-17 enol methylation led to clear distinctions of these isomeric products since the structures could no longer exist in enol-keto resonance. This allowed us to properly identify and characterize these reduced strictosidine aglycones and to identify demethylcorynantheidine (**2**) and demethyldihydrocorynantheine (**1**) end-products also generated by *C. roseus* enzymes CrGS and CrTHAS1 (Fig. 3a), which have not been described in previous studies. Based on the high production of demethyldihydrocorynantheine (**2**), we also suggest renaming CrTHAS1 as CrDCS.

In particular, the SABATH type methyltransferase MsC17OMT was able to methylate the C17-enol on a corynanthe skeleton, which was demonstrated by using the geissoschizine 19-epimers, and demethylcorynantheidine 20-epimers (Figs 3a, S4, S5). To our knowledge, this is the first example of enzymatic methylation of a nonaromatic enol in plant natural products. Other members of the SABATH methyltransferases included the founding members that methylate the carboxyl groups of plant hormones such as salicylic acid and jasmonic acid, as well as theobromine synthase that *N*-methylates 7-methylxanthine in coffee plants. The loganic acid methyltransferases (LAMTs) that performs carboxyl methylation of the MIA precursor loganic acid also belongs to this family (Fig. 2b). The identification and characterization of MsC17OMT therefore demonstrated a new and notable member in the SABATH methyltransferase family, which expands our understanding of natural product methylations.

We could only find C17OMT homologs in the plants from the Naucleeae tribe of the Rubiaceae family, such as *Uncaria rhynchophylla* (cat's claw), *Nauclea cadamba* (burflower tree), and *Cephalanthus occidentalis* (buttonbush). These plants naturally accumulate C17-methylated corynanthe MIAs and their simple derivatives such as the corresponding oxindoles and are not known to rearrange the corynanthe skeletons to complex MIA skeletons, such as strychnos, aspidosperma, or iboga types that are commonly found in MIA-producing plants in the Apocynaceae plants (Phillipson *et al.*, 1974; Martins & Nunez, 2015; Flores-Bocanegra *et al.*, 2020). *Catharanthus roseus* and *Rauvolfia serpentina*, both from the Apocynaceae family, can rearrange geissoschizine to form a strychnos (stemmadenine) or sarpagan (polyneuridine aldehyde) skeletons owing to the reactivity of the C17-enol and the cytochrome P450 monooxygenases that catalyze the intramolecular cyclization (Tatsis *et al.*, 2017; Qu *et al.*, 2018a). It is likely that the C17OMT evolved in these Naucleeae species to mask the reactive C17-enol, which is otherwise prone to nonspecific electrophilic addition to other cellular components, to reduce their cytotoxicity. However, such hypotheses require further experimental confirmation.

In a recent work (Schotte *et al.*, 2023), MsDCS1 and MsC17OMT were also discovered independently. This earlier

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study provided insights into the key residues involved in determining the corynanthe C20 stereochemistry between MsDCS1 (20*S* dominant) and another 20*R* dominant MsDCS, which has 64–68% amino acid identity to the MsDCS1-4 in this study. MsDCS1 and MsC17OMT function was also supported by their transient expression in *Nicotiana benthamiana* (tobacco) and co-infiltration of 4-methoxytryptamine and secologanin, leading to the formation of mitragynine.

While the 9-hydroxylase that completes mitragynine opioid biosynthesis awaits discovery, we further identified an alternative 9OMT from *Hamelia patens* (firebush). We demonstrated the Hp9OMT activity both *in vitro* and *in vivo* (Figs 3a, 5, S4–S7). While Hp9OMT has been useful to produce mitragynine in synthetic biology experiments, it is unlikely that Hp9OMT is involved in MIA 9-hydroxylation in firebush plant, since firebush naturally accumulates 10-hydroxylated heteroyohimbine type MIAs such as aricine and palmirine (Paniagua-Vega *et al.*, 2012) and is not known for producing either 9-hydroxylated MIAs or corynanthe type MIAs. Hp9OMT also accepted 10-hydroxycorynantheidine, its 20*R* isomer, and 10-hydroxygeissoschizine methyl ether 19-epimers, but not 11-hydroxylated versions of these MIAs as substrates (Fig. S5). Therefore, its activity is restricted to C9 and C10 on corynanthe MIAs. We hope to provide further updates on substrate specificity of Hp9OMT on heteroyohimbine MIA skeletons and the activity of other HpOMTs in future studies.

Interestingly, we could not identify a close homolog (> 50% amino acid identity) of Hp9OMT in the kratom transcriptomes (http://mpgr.uga.edu), which likely indicated that the Ms9OMT differs significantly from Hp9OMT in its primary sequence. In addition, 9OMT activity was not detected in assays using total kratom leaf protein extracts (Fig. 4). It is possible that the protein extraction method did not preserve 9OMT activity since we were also unable to detect the MsC17OMT activity from the same protein extract. It is also possible that the 9-hydroxylation and 9-*O*-methylation in kratom require different biosynthetic mechanism. Nonetheless, a bona fide Ms9OMT still needs to be discovered.

Finally, we successfully reconstituted the biosynthesis of kratom alkaloids mitragynine and speciogynine in both *E. coli* and yeast supplied with secologanin, tryptamine, and 4-hydroxytryptamine substrates. Both *in vitro* and *in vivo* studies confirmed our identifications of reaction products, which were either converted from known standards or converted enzymatically into known standards. Remarkably, the promiscuity of pathway enzymes allowed incorporation 4-hydroxytryptamine, which was generated by a fungal cytochrome P450 monooxygenase PcuT4H in yeast, into final alkaloid products. With enzyme promiscuity, we also showed the biosynthesis of unnatural MIAs such as 9-methoxygeissoschizine methyl ether epimers and 10-methoxycorynantheidine in *E. coli* (Figs S4, 5), which may also bear exciting opioid activities. It will be interesting to evaluate the opioid activity of 9-methoxygeissoschizine methyl ether epimers, since they are very similar to mitragynine except that the C20-19-18 vinyl side chain takes on a different confirmation. Based on the enzyme promiscuity demonstrated in our study, it is likely the remaining enzymes may tolerate or could be engineered to accept other indole substitutions for enzymatic production of kratom opioid derivatives for their pharmaceutical development. The identifications of the 9-hydroxylase and Ms9OMT will complete the mitragynine biosynthesis, which will also lead to their complete biosynthesis without the need for enzymes from other plant sources or organic synthesis.

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Competing interests

None declared.

Author contributions

KK, MS, VDL and YQ conceived and designed the research. KK, MS, JJOG-G, JEC, JG and YQ performed experiments and analyzed data. VDL and YQ wrote the manuscript. KK and MS contributed equally to this work.

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Data availability

The authors confirm that the data substantiating the findings of this study can be accessed within the article and its Supporting Information.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Dataset S1 Plant species, amino acid sequences, and Genbank numbers for the enzymes used in the phylogenetic study.

Fig. S1 SDS-PAGE image of His-tagged, purified recombinant proteins in this study.

Fig. S2 Electrospray Ionization tandem Mass Spectrometry fragment patterns of alkaloids involved in this study.

Fig. S3 Michaelis–Menten saturation kinetics of MsC17OMT.

Fig. S4 Liquid chromatography tandem mass spectrometry chromatograms of coupled enzyme activities from CrSTR, CrSGD, CrDCS (CrTHAS1), CpDCS, MsDCS1-4, MpDCS, CoDCS, MsC17OMT (MsOMT1), and Hp9OMT (HpOMT3), with secologanin and tryptamine/4-hydroxytryptamine substrates.

Fig. S5 Liquid chromatography tandem mass spectrometry chromatograms of the coupled enzyme activities from CrSTR, CrSGD, CrGS, CrDCS (CrTHAS1), CpDCS, MsDCS1-4, MpDCS, CoDCS, MsC17OMT (MsOMT1), and Hp9OMT (HpOMT3), with secologanin and 5-hydroxytryptamine/6-hydroxytryptamine substrates.

Fig. S6 Screening 9OMT activities from eight *Hamelia patens* HpOMT candidates *in vivo* in *Escherichia coli* with demethylmitragynine (10) and demethylspeciogynine (9) substrates.

Fig. S7 Mitragynine biosynthesis in *Escherichia coli* by co-expressing combinations of CrSTR, CrSGD, CrGS, MsDCS1, MsDCS4, MsC17OMT, and Hp9OMT.

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Table S1 Primers used in this study.

Table S2 Top expressing OMT candidates in *Hamelia patens* (firebush) leaf transcriptome.

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